PID/DATA MONITOR TABLE [BCM]

id0902f5996500

Note

- The [Type] in the PID/data monitor table is indicated as follows.
  - — Input: Value input to module
  - — Conversion: Value converted into value input to module
  - — Calculation: Value calculated for several input values in module
  - — Output: Value output by module

—: Not applicable

Item	Unit	Type	Content	Operation status (reference)	Terminal
ABK_HD_CS_R	If an accessory part is installed, it cannot be used.				
ABK_HN_CS_R	If an accessory part is installed, it cannot be used.				
BG_HORN_RLY	If an accessory part is installed, it cannot be used.				
BLWR_RLY	Off/On	Output	Displays the blower relay operation status.	• Off: Blower relay is off (ignition is switched off) • On: Blower relay is on (ignition is switched ON)	3E
BRAKE_SW	Off/On	Conversion	Displays the brake switch (No.1 signal) operation status.	• Off: Brake switch (No. 1 signal) is off (brake pedal is not depressed) • On: Brake switch (No. 1 signal) is on (brake pedal is depressed)	2Z
BRK_F_L_SW	Off(Normal)/On (Low)	Conversion	Displays the brake fluid level status.	• Off (Normal): Brake fluid level is between MAX and MIN • On (Low): Brake fluid level is lower than MIN	3R
C_DL_SW_LK	Off/On	Conversion	Displays the door lock switch operation status (lock operation).	• Off: door lock switch is not in lock • On: door lock switch is in lock	2K
C_DL_SW_UNL	Off/On	Conversion	Displays the door lock switch operation status (unlock operation).	• Off: door lock switch is not in unlock • On: door lock switch is in unlock	2K
DOOR_ALL	Close/OPEN	Conversion	Displays all the doors (front door, rear door) open/ closed condition.	• Close: All doors are closed • OPEN: Any door is open	2H/2G/ 2F/2E
DOOR_D	Close/OPEN	Conversion	Displays the front door (RH) open/closed condition.	• Close: Front door (RH) is closed • OPEN: Front door (RH) is open	2H
DOOR_LR	Close/OPEN	Conversion	Displays the rear door (LH) open/closed condition.	• Close: Rear door (LH) is closed • OPEN: Rear door (LH) is open	2E
DOOR_P	Close/OPEN	Conversion	Displays the front door (LH) open/closed condition.	• Close: Front door (LH) is closed • OPEN: Front door (LH) is open	2G
DOOR_RR	Close/OPEN	Conversion	Displays the rear door (RH) open/closed condition.	• Close: Rear door (RH) is closed • OPEN: Rear door (RH) is open	2F
DR_LK_RY	Off/On	Output	Displays the door lock relay operation request (lock signal) send status. The BCM sends a lock signal to the ESU via LIN communication.	• Off: Lock signal is not sent (lock operation is not operated) • On: Lock signal is sent (lock operation is operated)	2Y

Item	Unit	Type	Content	Operation status (reference)	Terminal
DR_UNLK_RY	Off/On	Output	Displays the door unlock relay operation request (unlock signal) send status. The BCM sends an unlock signal to the ESU via LIN communication.	• Off: Unlock signal is not sent (unlock operation is not operated) • On: Unlock signal is sent (unlock operation is performed)	2Y
F_FOG_RELAY	Off/On	Output	Displays the front fog light relay operation status.	• Off: Front fog light relay is off (fog light switch is in OFF position) • On: Front fog light relay is on (light switch is in TNS or HEAD position and fog light switch is in F.FOG position)	3I
F_FOG_SW	Off/On	Conversion	Displays the fog light switch operation status. The BCM receives a fog light switch input signal via CAN communication from the start stop unit.	• Off: Fog light switch OFF signal is received (fog light switch is in OFF position) • On: Fog light switch ON signal is received (fog light switch is in F.FOG position)	2C/2D
FUEL_SEN_HM	Displayed on M-MDS but does not function.				
FUEL_SEN_HS	Displayed on M-MDS but does not function.				
FUEL_SEN_HV	Displayed on M-MDS but does not function.				
FUEL_SEN_M	V	Input	Displays the fuel gauge sender unit input voltage. If the fuel tank level is low, the BCM input voltage increases because the fuel gauge sender unit resistance increases.	Displays the fuel gauge sender unit voltage.	2Q
FUEL_SEN_S	Displayed on M-MDS but does not function.				
FUEL_SEN_V	V	Input	Displays the fuel gauge sender unit supply voltage.	Ignition switched ON (engine off): Approx. 4.15 V	1A/1B
H/L_CLN_RY	Off/On	Output	Displays the headlight cleaner relay operation status.	**Note** • For the specific operation condition of the PID [H/L_CLN_RY], refer to the [HEADLIGHT CLEANER SYSTEM] for new vehicle instruction. • Off: Headlight cleaner relay is off (headlight cleaner system is not operating) • On: Headlight cleaner relay is on (headlight cleaner system is operating)	3J
H/L_CS_RLS	Off/On	Conversion	Displays the headlight LO on request signal reception status. The BCM receives a headlight LO on request signal via LIN communication from the auto-light sensor/rain sensor.	**Note** • When the auto-light sensor/rain sensor detects the average 400 lux or less per 1 min for upward illumination, it sends a headlight LO on request signal to the BCM. • Off: Headlight LO on request signal is not received • On: Headlight LO on request signal is received	3X

Item	Unit	Type	Content	Operation status (reference)	Terminal
H/L_DRL	Off/On	Output	Displays the running light on request signal send status. The BCM sends a running light on request signal to the ESU via LIN communication.	**Note** • For the specific operation condition of the PID [H/L_DRL], refer to the [RUNNING LIGHT SYSTEM] for new vehicle instruction. • Off: Running light on request signal is not sent • On: Running light on request signal is sent	2Y
H/L_HI	%	Output	Displays the headlight HI on request signal send status. The BCM sends a headlight HI on request signal to the ESU via LIN communication.	• Headlight HI on request signal is not sent (headlight high beam turns off): 0 % • Headlight HI on request signal is sent (headlight high beam turns on): 100 %	2Y
H/L_LOW	%	Output	Displays the headlight LO on request signal send status. The BCM sends a headlight LO on request signal to the ESU via LIN communication.	• Headlight LO on request signal is not sent (headlight low beam turns off): 0 % • Headlight LO on request signal is sent (headlight low beam turns on): 100 %	2Y
H/L_PASS	Off/On	Output	Displays the headlight LO and HI on request signal send status. The BCM sends a headlight LO and HI on request signal to the ESU via LIN communication.	• Off: Headlight LO and HI on request signal is not sent (no passing) • On: Headlight LO and HI on request signal is sent (passing)	2Y
H/L_SW_HI	Off/On	Conversion	Displays the light switch (HI position) operation status. The BCM receives a light switch (HI position) input signal via CAN communication from the start stop unit.	• Off: Light switch (HI position) OFF signal is received • On: Light switch (HI position) ON signal is received	2C/2D
H/L_SW_LOW	Off/On	Conversion	Displays the light switch (LO position) operation status. The BCM receives a light switch (LO position) input signal via CAN communication from the start stop unit.	• Off: Light switch (LO position) off signal is received • On: Light switch (LO position) on signal is received	2C/2D
H/L_SW_OFF	Off/On	Conversion	Displays the light switch (OFF position) operation status. The BCM receives a light switch (OFF position) input signal via CAN communication from the start stop unit.	• Off: Light switch (OFF position) off signal is received • On: Light switch (OFF position) on signal is received	2C/2D
H/L_SW_PASS	Off/On	Conversion	Displays the light switch (passing position) operation status. The BCM receives a light switch (passing position) input signal via CAN communication from the start stop unit.	• Off: Light switch (passing position) off signal is received • On: Light switch (passing position) on signal is received	2C/2D

Item	Unit	Type	Content	Operation status (reference)	Terminal
H/L_SW_TNS	Off/On	Conversion	Displays the light switch (TNS position) operation status. The BCM receives a light switch (TNS position) input signal via CAN communication from the start stop unit.	• Off: Light switch (TNS position) off signal is received • On: Light switch (TNS position) on signal is received	2C/2D
H/L_TNS	Off/Right-On/Left-On/On	Output	Displays the TNS on request signal send status. The BCM sends a TNS on request signal to the ESU via LIN communication.	• Off: TNS on request signal is not sent (light switch is in position other than TNS) • Right-On: Parking light (RH) is forcibly operated using simulation item • Left-On: Parking light (LH) is forcibly operated using simulation item • On: TNS on request signal is sent (light switch is in TNS position)	2Y
H/L_TNS_RLS	Off/On	Conversion	Displays the TNS on request signal reception status. The BCM receives a TNS on request signal via LIN communication from the auto-light sensor/rain sensor.	**Note** • When the auto-light sensor/rain sensor detects the average 700 lux or less per 1 min for upward illumination, it sends a TNS on request signal to the BCM. • Off: TNS on request signal is not received • On: TNS on request signal is received	3X
H/L_TYPE	Halogen_2Lights/ Halogen_4Lights/ Lo-HID&Hi-Hlogen_4Lights/ LED_4Lights/Bi-Xenon	—	Displays the type of headlight equipped to the vehicle. The BCM sends information concerning the type of headlight to the ESU via LIN communication.	—	2Y
HAZARD_SW	Off/On/Unknown	Conversion	Displays the hazard warning operation status. The BCM receives a hazard warning switch input signal via CAN communication from the start stop unit.	• Off: Hazard warning switch off signal is received (hazard warning switch is not pressed) • On: Hazard warning switch on signal is received (hazard warning switch is pressed) • Unknown: Signal which hazard warning switch position is not determined is received	2C/2D
HOOD	Close/OPEN	Conversion	Displays the bonnet latch switch operation status.	• Close: Bonnet latch switch is on (bonnet is closed) • OPEN: Bonnet latch switch is off (bonnet is open)	3W
HORN_RLY	Off/On	Output	Displays the horn relay operation status.	• Off: Horn relay is off (horn switch is not pressed) • On: Horn relay is on (horn switch is pressed)	3H
IG1_STATUS	IG-Off/IG-On	Conversion	Displays the IG1 power supply input status determined in the BCM.	• IG-Off: IG1 power is not input to BCM • IG-On: IG1 power is input to BCM	3O

Item	Unit	Type	Content	Operation status (reference)	Terminal
KC_SW_D_LK	Off/On	Conversion	Displays the front door key cylinder switch operation status (lock operation).	• Off: Front door key cylinder switch is not in lock (lock operation is not performed using front door key cylinder) • On: Front door key cylinder switch is in lock (lock operation is performed using front door key cylinder)	2L
KC_SW_D_UNL	Off/On	Conversion	Displays the front door key cylinder switch operation status (unlock operation).	• Off: Front door key cylinder switch is not in unlock (unlock operation is not performed using front door key cylinder) • On: Front door key cylinder switch is in unlock (unlock operation is performed using front door key cylinder)	2L
LL_SW_D_LK	Off/On	Conversion	Displays the front door lock-link switch operation status (lock operation).	• Off: Front door lock-link switch is not in lock (lock operation is not performed using driver's door lock knob) • On: Front door lock-link switch is in lock (lock operation is performed using driver's door lock knob)	2A
LL_SW_D_UNL	Off/On	Conversion	Displays the front door lock-link switch operation status (unlock operation).	• Off: Front door lock-link switch is not in unlock (unlock operation is not performed using driver's door lock knob) • On: Front door lock-link switch is in unlock (unlock operation is performed using driver's door lock knob)	2B
LL_SW_P/R	Off/On	Conversion	Displays the door lock-link switch (passenger-side/ rear-side) operation status (lock operation).	• Off: Door lock-link switch (passenger-side/rear-side) is not in lock (lock operation is not performed using door lock knob (passenger-side/rear-side)) • On: Door lock-link switch (passenger-side/rear-side) is in lock (lock operation is performed using door lock knob (passenger-side/rear-side))	2M
PNC_HD_CS_R	If an accessory part is installed, it cannot be used.				
PNC_HN_CS_R	If an accessory part is installed, it cannot be used.				
PTC_HEAT_ST	%	Output	Displays the PTC heater operation status (PTC heater output value controlled by BCM).	—	2AA
R_DEF_RLY	Off/On	Output	Displays the rear window defogger relay operation status.	• Off: Rear window defogger relay is off (rear window defogger switch is not pressed) • On: Rear window defogger relay is on (rear window defogger switch is pressed)	3K

Item	Unit	Type	Content	Operation status (reference)	Terminal
R_FOG_LAMP	Off/On	Output	Displays the rear fog light on request signal send status. The BCM sends a rear fog light on request signal to the ESU via LIN communication.	• Off: Rear fog light on request signal is not sent (fog light switch is in OFF position) • On: Rear fog light on request signal is sent (fog light switch is in TNS or HEAD position and fog light switch is in R.FOG position)	2Y
R_FOG_SW	Off/On	Conversion	Displays the fog light switch operation status. The BCM receives a fog light switch input signal via CAN communication from the start stop unit.	• Off: Fog light switch OFF signal is received (fog light switch is in OFF position) • On: Fog light switch ON signal is received (fog light switch is in R.FOG position)	2C/2D
REVERSE_LMP	Off/On	Output	Displays the back-up light on request signal send status. The BCM sends a back-up light on request signal to the ESU via LIN communication.	• Off: Back-up light on request signal is not sent (MTX: shift lever is not in reverse position, ATX: selector lever is not in R position) • On: Back-up light on request signal is sent (MTX: shift lever is in reverse position, ATX: selector lever is in R position)	2Y
RLS_AAT	Displayed on M-MDS but does not function.				
RLS_WS_TEMP	Displayed on M-MDS but does not function.				
ROOM_LAMP	Off/100%_On/60%_On/50%_On/40%_On/30%_On/20%_On	Output	Displays the front map light on request signal send status. The BCM sends a front map light on request signal to the ESU via LIN communication.	**Note** • For the specific operation condition of the PID [ROOM_LAMP], refer to the [INTERIOR LIGHT CONTROL SYSTEM] for new vehicle instruction. • Off: Front map light on request signal is not sent • 100%_On: Front map light on request signal (100%) is sent • 60%_On: Front map light on request signal (60%) is sent • 50%_On:— • 40%_On:— • 30%_On:— • 20%_On:—	2Y
SEAT_B_2C	Unbuckled/Buckled	Conversion	Displays the rear center buckle switch operation status.	• Unbuckled: Rear center buckle switch is off (rear seat belt (center) is not fastened) • Buckled: Rear center buckle switch is on (rear seat belt (center) is fastened)	2O
SEAT_B_2L	Unbuckled/Buckled	Conversion	Displays the rear buckle switch (LH) operation status.	• Unbuckled: Rear buckle switch (LH) is off (rear seat belt (LH) is not fastened) • Buckled: Rear buckle switch (LH) is on (rear seat belt (LH) is fastened)	2N
SEAT_B_2R	Unbuckled/Buckled	Conversion	Displays the rear buckle switch (RH) operation status.	• Unbuckled: Rear buckle switch (RH) is off (rear seat belt (RH) is not fastened) • Buckled: Rear buckle switch (RH) is on (rear seat belt (RH) is fastened)	2P
SEAT_B_3C	Displayed on M-MDS but does not function.				
SEAT_B_3L	Displayed on M-MDS but does not function.				
SEAT_B_3R	Displayed on M-MDS but does not function.				

Item	Unit	Type	Content	Operation status (reference)	Terminal
STOP_LAMP_DSC	Unknown/Off/On/ Backup_On	Conversion	Displays the brake light signal reception status. The BCM receives a brake light signal via CAN communication from the DSC HU/CM.	• Unknown: Brake light signal cannot be received • Off: Brake light signal (turn-off) is received (SCBS brake control is not operating) • On: Brake light signal (turn-on) is received (SCBS brake control is operating) • Backup_On: Brake light signal (turn-on) is received (malfunction is occurring in brake light switch)	3C/3D
STOP_LMP_CTL	Off/On	Output	Displays the brake light relay operation status.	**Note** • When the brake control (SCBS brake) is operating, the brake light relay turns on. • Off: Brake light relay is off (brake pedal is not depressed) • On: Brake light relay is on (brake pedal is depressed)	1D
STOP_LMP_IN	Off/On	Conversion	Displays the brake light operation status.	• Off: Brake light turns off (brake pedal is not depressed) • On: Brake light turns on (brake pedal is depressed)	2AB
TR_UNL_SW	Off/On	Conversion	**4SD** • Displays the trunk lid opener switch (exterior) operation status. **5HB** • Displays the liftgate opener switch operation status.	**4SD** • Off: Trunk lid opener switch (exterior) is off (trunk lid opener switch (exterior) is not pressed) • On: Trunk lid opener switch (exterior) is on (trunk lid opener switch (exterior) is pressed) **5HB** • Off: Liftgate opener switch is off (liftgate opener switch is not pressed) • On: Liftgate opener switch is on (liftgate opener switch is pressed)	2S
TR_UNL_SW_IP	Off/On	Conversion	Displays the trunk lid opener switch (interior) operation status.	• Off: Trunk lid opener switch (interior) is off (trunk lid opener switch (interior) is not pressed) • On: Trunk lid opener switch (interior) is on (trunk lid opener switch (interior) is pressed)	3U
TRUNK	Close/OPEN	Conversion	**4SD** • Displays the trunk lid latch switch operation status. **5HB** • Displays the Liftgate latch switch operation status.	**4SD** • Close: Trunk lid latch switch is off (trunk is closed) • OPEN: Trunk lid latch switch is on (trunk is open) **5HB** • Close: Liftgate latch switch is off (liftgate is closed) • OPEN: Liftgate latch switch is on (liftgate is open)	2AC

Item	Unit	Type	Content	Operation status (reference)	Terminal
TRUNK_LAMP	Off/On	Output	**4SD** • Displays the trunk compartment light on request signal send status. The BCM sends a trunk compartment light on request signal to the ESU via LIN communication. **5HB** • Displays the cargo compartment light on request signal send status. The BCM sends a cargo compartment light on request signal to the ESU via LIN communication.	**4SD** • Off: Trunk compartment light on request signal is not sent (trunk is closed) • On: Trunk compartment light on request signal is sent (trunk is open) **5HB** • Off: Cargo compartment light on request signal is not sent (liftgate is closed) • On: Cargo compartment light on request signal is sent (liftgate is open)	2Y
TRUNK_MOTOR	Off/On	Output	**4SD** • Displays the trunk release signal send status. The BCM sends a trunk release signal to the ESU via LIN communication. **5HB** • Displays the liftgate release signal send status. The BCM sends a liftgate release signal to the ESU via LIN communication.	**4SD** • Off: Trunk release signal is not sent (trunk lid opener switch (interior) and trunk lid opener switch(exterior) is not pressed) • On: Trunk release signal is sent (trunk lid opener switch (interior) or trunk lid opener switch(exterior) is pressed) **5HB** • Off: Liftgate release signal is not sent (liftgate opener switch is not pressed) • On: Liftgate release signal is sent (liftgate opener switch is pressed)	2Y
TURN_L_L	Off/On	Output	Displays the turn light (LH) operation status.	• Off: Turn light (LH) turns off (turn switch is in OFF position) • On: Turn light (LH) flashes (turn switch is in LH position)	1G/3B
TURN_L_R	Off/On	Output	Displays the turn light (RH) operation status.	• Off: Turn light (RH) turns off (turn switch is in OFF position) • On: Turn light (RH) flashes (turn switch is in RH position)	1E/3A
TURN_SW	Off/Turn_R_On/ Turn_L_On/ Unknown	Conversion	Displays the turn switch operation status. The BCM receives a turn switch input signal via CAN communication from the start stop unit.	• Off: Turn switch off signal is received (turn switch is in OFF position) • Turn_R_On: Turn switch RH signal is received (turn switch is in RH position) • Turn_L_On: Turn switch LH signal is received (turn switch is in LH position) • Unknown: Signal which turn switch position is not determined is received	2C/2D
VPWR	V	Input	Displays the BCM supply voltage.	Switch the ignition ON (engine off):Approx. 12 V	1A
VPWR_B_H	V	Input	Displays the BCM supply voltage.	Switch the ignition ON (engine off):Approx. 12 V	1A
VSPD	KPH	Output	Displays the vehicle speed. The BCM sends a vehicle speed signal to the auto-light sensor/rain sensor via LIN communication.	—	3X

Item	Unit	Type	Content	Operation status (reference)	Terminal
WASHER_CTRL	Off/Front/Rear	Output	Displays the washer operation request signal send status. The BCM sends a washer operation request signal to the ESU via LIN communication.	• Off: Washer operation request signal is not sent (washer is not operating) • Front: Front washer operation request signal is sent (front washer switch is in ON position) • Rear: Rear washer operation request signal is sent (rear washer switch is in rear washer position)	2Y
WASHER_F_SW	Off/On	Conversion	Displays the front washer switch operation status. The BCM receives a front washer switch input signal via CAN communication from the start stop unit.	• Off: Front washer switch off signal is received (front washer switch is in OFF position) • On: Front washer switch on signal is received (front washer switch is in ON position)	2C/2D
WASHER_FLD	Off(Normal)/On (Low)	Conversion	Displays the washer fluid level status.	• Off (Normal): Washer fluid level is normal • On (Low): Washer fluid level is low	3Q
WASHER_R_SW	Off/On	Conversion	Displays the rear washer switch operation status. The BCM receives a rear washer switch input signal via CAN communication from the start stop unit.	• Off: Rear washer switch off signal is received (rear washer switch is in OFF position) • On: Rear washer switch on signal is received (rear washer switch is in rear washer position)	2C/2D
WIP_CS_RLS	Off/Single_Wipe/ SPEED_LOW/ SPEED_HI/Failure	Conversion	Displays the auto wiper control request signal reception status. The BCM receives an auto wiper control request signal via LIN communication from the auto-light sensor/rain sensor.	• Off: Off signal is received • Single_Wipe: Intermittent operation signal is received • SPEED_LOW: Continuous low operation signal is received • SPEED_HI: High speed signal is received • Failure: Auto wiper control request signal cannot be received	3X
WIP_F_AS	P_Posi/Not_P_Posi	Conversion	Displays the autostop switch operation status (windshield wiper arm and blade position).	• P_Posi: Windshield wipers are stopped in park position • Not_P_Posi: Windshield wipers are stopped in position other than park	3M
WIP_F_INT_L	%	Conversion	Displays the sensitivity adjustment volume operation status. The BCM receives a sensitivity adjustment volume input signal via CAN communication from the start stop unit.	The value changes according to the sensitivity adjustment volume operation.	2C/2D
WIP_F_RY	Off/On	Output	Displays the wiper INT relay operation status.	• Off: Wiper INT relay is off (windshield wiper are not operating) • On: Wiper INT relay is on (windshield wipers are operating)	3F

Item	Unit	Type	Content	Operation status (reference)	Terminal
WIP_F_RY_HI	Off/On	Output	Displays the wiper HI relay operation status.	**Note** • If the windshield wiper switch is in the INT/AUTO position, the wiper HI relay turns on depending on the amount of rainfall. • Off: Wiper HI relay is off (windshield wiper switch is in position other than HI) • On: Wiper HI relay is on (windshield wiper switch is in HI position)	3G
WIP_F_SW	OFF/LOW/HI/INT / AUTO	Conversion	Displays the windshield wiper switch operation status. The BCM receives a windshield wiper switch input signal via CAN communication from the start stop unit.	• Off: Windshield wiper switch off signal is received (windshield wiper switch is in OFF position) • LOW: Windshield wiper switch LOW signal is received (windshield wiper switch is in LO position) • HI: Windshield wiper switch HI signal is received (windshield wiper switch is in HI position) • INT/AUTO: Windshield wiper switch INT signal or AUTO signal is received (windshield wiper switch is in INT/AUTO position)	2C/2D
WIP_R_CTRL	Off/INT/On/ On_1Time/ On_2Time/ On_3Time/ On_4Time	Output	Displays the rear wiper operation request signal send status. The BCM sends a rear wiper operation request signal to the ESU via LIN communication.	• Off: Rear wiper operation request signal is not sent (rear wiper switch is in OFF position) • INT: Rear wiper INT operation request signal is sent (rear wiper switch is in INT position) • On: Rear wiper operation request signal is sent (rear wiper switch is in ON position) • On_1Time:— • On_2Time: Rear wiper INT operation request signal is sent (during synchronized washer operation) • On_3Time:— • On_4Time:—	2Y
WIP_R_INT	s	Output	Displays the rear wiper intermittent interval. The BCM sends a rear wiper intermittent interval to the ESU via LIN communication.	—	2Y
WIP_R_SW	OFF/LOW/INT	Conversion	Displays the rear wiper switch operation status. The BCM receives a rear wiper switch input signal via CAN communication from the start stop unit.	• Off: Rear wiper switch off signal is received (rear wiper switch is in OFF position) • LOW: Rear wiper switch LOW signal is received (rear wiper switch is in ON position) • INT: Rear wiper switch INT signal is received (rear wiper switch is in INT position)	2C/2D